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Composed Sketch-Text Visual Grounding in Remote Sensing via Progressive Cross-Scale Fusion

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***Abstract*—Visual grounding in remote sensing aims to localize target objects in satellite imagery based on textual queries. However, complex spatial relationships and object geometries are often difficult to describe concisely using natural language alone. To address this limitation, we investigate composed sketch-text visual grounding, which jointly uses text descriptions and sketches as complementary queries for target localization. Sketches complement textual descriptions by conveying spatial configurations and geometric cues that are cumbersome to express verbally, thereby enabling more concise and informative queries. To tackle this task, we propose CSTVG, a novel framework comprising a multimodal encoder and a progressive cross-scale fusion module. Built upon the pre-trained CLIP model, the multimodal encoder incorporates a cross-modal interaction module to effectively align and integrate textual and sketch representations. The progressive cross-scale fusion module subsequently refines the multimodal query representations and integrates them with visual features extracted from satellite images at multiple resolutions, producing unified features for target localization. Extensive experiments on CRSVG show that CSTVG outperforms existing state-of-the-art methods and that combining text and sketches substantially improves grounding accuracy over conventional text-only visual grounding approaches. The code is available at https://github.com/shiyihua123/CSTVG.**

***Index Terms*—Composed visual grounding, sketches, multimodal retrieval, cross-modal interaction.**

I. Introduction

VISUAL grounding in remote sensing [1], [2] aims to locate an object of interest (in the form of a bounding box or segmentation mask) from a satellite image based on a text query. Satellite images typically cover a broad area of the Earth's surface and contain numerous ground objects. Visual grounding can quickly and accurately locate a ground object, which has great potential in various tasks, such as disaster management [3], drone navigation [4], and search and rescue [5], [6].

Satellite images often contain various objects that are difficult to distinguish from an overhead perspective. When short text queries are used to locate an object, there is a high risk of retrieving incorrect or ambiguous results. For example, in Fig. 1(a), on the left, a short query attempts to locate a storage tank, but the image contains several storage tanks, making it challenging to pinpoint the correct one. The lack of specificity in the query hampers the identification of a unique target object. To enhance accuracy, it is crucial to incorporate spatial relationships between multiple reference objects. However, relying solely on text can be cumbersome, especially for complex shapes or multiple objects [7], [8]. As shown in Fig. 1(a), on the right, a longer query specifies a particular storage tank by describing not only the tank but also a building, a pond, their respective areas, distances, and spatial relationships. This extended description helps narrow down the search and defines the target object more precisely. While this extended description helps uniquely identify the target, it also adds complexity. The query needs to describe the unique attributes and spatial relationships of each object, and users must carefully articulate these details. Such lengthy queries can be challenging for users to construct and for the model to process, which increases the likelihood of misidentification. Thus, relying solely on text descriptions to locate objects in satellite images may lead to inefficiencies and reduced accuracy.

Sketches can effectively illustrate the spatial relationships and contours of objects [9]–[11]. As demonstrated in Fig. 1(b), the sketch clearly shows the relationship between the storage tank and other objects. The contours and orientations of each object also serve as useful search cues. However, sketches alone are insufficient for precisely representing query requirements, as different object categories may have similar contour lines from a top-down view. For example, buildings and tennis courts can exhibit similar contours, making it challenging to differentiate between them based solely on sketches. In addition, satellite images provide geographical and attribute information that sketches cannot convey, such as numerical geographical details, distances, and object colors. Therefore, to accurately and clearly express query requirements, it is essential to combine text with sketches for object queries. To address this, we propose a composed visual grounding task for remote sensing images.

Composed visual grounding needs to handle a greater variety of input modalities, such as text, sketches, and satellite images, introducing several new challenges. First, each composed query integrates data from multiple modalities, complicating its understanding due to the inherent differences between these modalities. To effectively understand and extract features from

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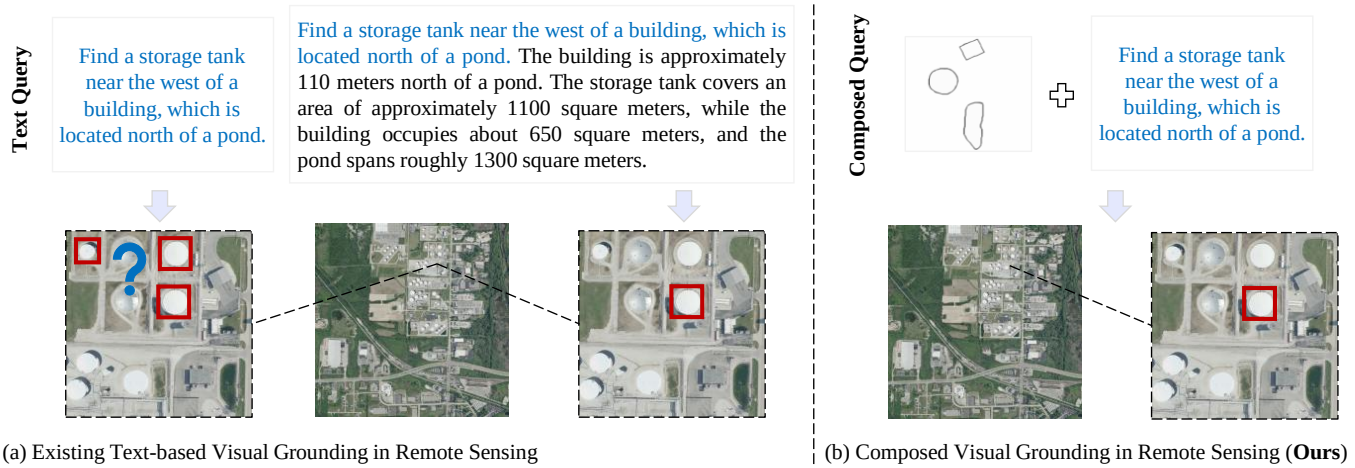
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(a) Existing Text-based Visual Grounding in Remote Sensing (b) Composed Visual Grounding in Remote Sensing (**Ours**)

Fig. 1. Comparison of existing text-based visual grounding methods and our composed visual grounding method.

these queries, it is crucial to align and bridge the semantic gaps between the various data modalities. Second, the presence of diverse modalities in composed queries complicates the feature fusion process. Typically, visual grounding involves extracting features from both the query and the reference, followed by fusing these features to create a grounding head for object prediction. However, the introduction of more complex modalities in composed queries makes this fusion process more challenging and can affect the accuracy of object localization.

To address these challenges, we propose a novel framework called CSTVG for composed sketch-text visual grounding in remote sensing. CSTVG includes a multimodal encoder and a progressive cross-scale fusion module. The multimodal encoder utilizes a pre-trained CLIP model [12] and incorporates a cross-modal interaction module to align and narrow the semantic gap between text and sketches. The fusion module progressively refines the composed query features by integrating them with reference features at multiple resolutions. This approach mimics the way humans adjust their search strategies on a map, focusing on varying levels of detail based on the content observed. By gradually merging query features with reference features, the fusion module effectively generates a grounding head, leading to more precise object predictions.

To support the evaluation of composed visual grounding methods, we use CRSVG, a large-scale benchmark dataset first introduced in our preliminary short conference paper [13]. Constructed with the assistance of OpenStreetMap, CRSVG comprises remote sensing images, query texts, sketches, and target bounding boxes. We further supplement the dataset with human-drawn sketches for robustness evaluation and provide more detailed descriptions of its construction process and statistical analyses. Furthermore, we extend our preliminary work by introducing a novel progressive cross-scale fusion module for more effective query-guided multi-scale feature integration, along with more comprehensive experiments and analyses. Extensive experiments on CRSVG demonstrate that CSTVG consistently outperforms state-of-the-art methods.

We summarize the main contributions of this work as follows:

- We further investigate composed sketch-text visual grounding in remote sensing, provide a more detailed description and statistical analysis of CRSVG, and supplement the dataset with additional human-drawn sketches for robustness evaluation.
- We propose a novel framework, CSTVG, which integrates a multimodal encoder with a progressive cross-scale fusion module to effectively represent and fuse multimodal features, thereby improving object localization accuracy.
- We conduct extensive experiments on CRSVG, including comparisons with state-of-the-art methods, qualitative analyses, and ablation studies. The results demonstrate the superior localization performance of our proposed CSTVG and the effectiveness of combining text and sketches.

The rest of this paper is organized as follows. Section II provides a brief review of related work on visual grounding. Section III describes the problem definition and dataset construction. Section IV presents a detailed description of our method. In Section V, we present the results of our comprehensive experiments on composed visual grounding. Finally, we conclude the paper with some remarks in Section VI.

II. Related Work

A. Visual Grounding Datasets

We provide an overview of existing 2D visual grounding datasets for both natural and remote sensing images. The RefCOCO [14], RefCOCO+ [14], and RefCOCOg [15] datasets, based on MSCOCO [16] images, use natural images with single expressions for object localization. The GuessWhat?! [17] dataset, also derived from MSCOCO images, employs a dialogue-based approach for visual grounding but evaluates only the final results of models, without assessing their reasoning processes. CLEVR-Ref+ [18], a synthetic diagnostic dataset, is designed to evaluate intermediate reasoning processes; however, its synthetic nature limits its applicability to real-world scenarios. Cops-Ref [19] utilizes real natural images to explore complete reasoning chains expressed in language. KB-Ref [20] incorporates commonsense knowledge

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from Wikipedia and ConceptNet [21], though its focus is on small-scale scenes. GigaGrounding [22] introduces a large-scale, gigapixel-level dataset for visual grounding, primarily focusing on natural images. For remote sensing images covering extensive areas, RSVG [2] uses text templates to create a visual grounding dataset in the remote sensing domain, while RSVGD [1] constructs a larger visual grounding dataset. GeoText-1652 [4] uses drone-view images to develop a natural language-guided geolocalization benchmark. Notably, all these datasets rely solely on text and do not address the use of composed queries for visual grounding.

B. Visual Grounding Methods

Visual grounding in remote sensing predominantly relies on one-stage architectures, which directly regress bounding boxes from input queries without the computational overhead of region proposal networks [2], [23]. This paradigm is favored for its efficiency and end-to-end trainability. We review the evolution of one-stage techniques, categorizing them into methods developed for natural images and those specifically tailored for remote sensing domains.

a) ***Methods for Natural Imagery****:* Early one-stage approaches focused on efficient feature fusion. For example, FAOA [24] encodes linguistic expressions into a single global vector and concatenates it across all spatial locations of visual features for localization. However, this approach struggles with longer expressions and often loses important details. To address this limitation, ReSC [25] introduced a recursive sub-query construction framework that decomposes complex sentences into multiple sub-queries, iteratively refining alignment. Nevertheless, ReSC primarily focuses on semantic decomposition and lacks explicit modeling of spatial relationships between objects.

To incorporate spatial context, LBYLNet [26] employs landmark feature convolution to encode contextual cues within the image. Subsequently, transformer-based architectures have gained prominence. For example, TransVG [27] leverages self-attention mechanisms to capture long-range dependencies between text and image patches, while VLTVG [28] enhances this with a visual-linguistic verification module to generate text-conditioned discriminative features. These methods rely on labeled data, which is costly to collect. To overcome this, GroundVLP [29] explored zero-shot generalization by integrating GradCAM-generated heatmaps [30] with open-vocabulary object proposals. Despite these advances, the aforementioned methods are predominantly designed for natural images and may suffer performance degradation when applied to remote sensing images, which differ significantly from natural images.

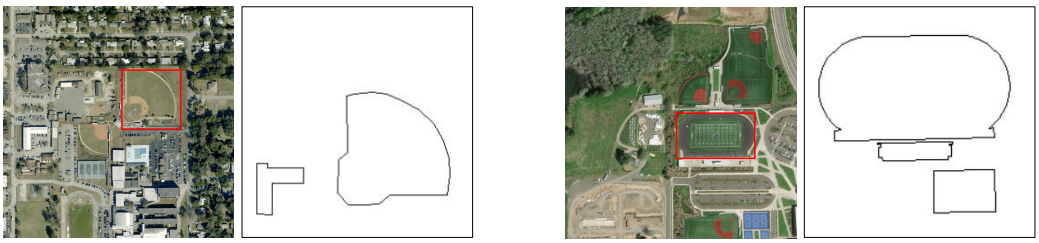
b) ***Methods for Remote Sensing Imagery****:* Recent studies have advanced visual grounding in remote sensing images. For example, GeoVG [2] incorporates numerical geospatial relations into the language encoder, enabling the model to interpret directional and metric constraints alongside visual features. To handle dense and cluttered aerial scenes, RINet [31] explicitly models spatial dependencies between candidate regions and their surroundings, significantly improving localization accuracy. Further refining cross-modal interaction, LQVG [32] uses a language query-based transformer to align textual semantics with satellite imagery through deep fusion layers, while LPVA [33] employs a language-guided progressive visual attention mechanism that dynamically updates visual features based on linguistic cues. Similarly, QAMFN [34] integrates attention mechanisms directly into the visual encoder to enhance early-stage text–image alignment. Recent studies have further extended remote sensing visual grounding to more challenging settings. TCSG [35] introduces text-conditioned spatial priors and sparse sampling to improve referential modeling and suppress background interference. DroneVG-R1 [36] employs an Implicit-to-Explicit Chain-of-Thought strategy to resolve implicit references in UAV imagery. MIGRANT [37] extends grounding to multiple images through hierarchical grounding data and two-stage instruction tuning. Despite these advances, these methods still rely primarily on language queries for grounding.

Beyond task-specific grounding networks, recent advances in remote sensing vision-language foundation models have provided more transferable representations for multimodal remote sensing understanding. RemoteCLIP [38] adapts contrastive vision-language pretraining to remote sensing imagery, learning semantically aligned visual and textual representations for downstream tasks. Dif-Fusion [39] leverages diffusion-based generative mechanisms to effectively aggregate multi-source information, establishing a new paradigm for high-fidelity, high-resolution image acquisition and facilitating high-precision visual analysis and localization. RS5M/GeoRSCLIP [40] further scale vision-language pretraining by constructing a large-scale remote sensing image-text dataset and developing a domain-specific vision-language model. GRAFT [41] explores annotation-free vision-language pretraining by aligning satellite imagery with co-located ground-level images. Together, these efforts improve the transferability and generalization of vision-language representations, providing a stronger foundation for downstream remote sensing understanding and grounding tasks.

More recently, open-vocabulary and foundation model paradigms have been explored to improve generalization. RSGPT [42], for instance, adapts pre-trained MLLMs for image captioning and visual question answering, focusing on image-level understanding but lacking fine-grained localization. Models such as GeoChat [43], SkyEyeGPT [44], EarthGPT [45], LHRS-Bot [46], and EarthMarker [47] extend these capabilities to the region level, enabling more precise visual grounding in complex scenes. For example, EarthGPT further introduces a multimodal large language model tailored for remote sensing, strengthening cross-modal understanding. RSVG-ZeroOV [48] demonstrates the potential of frozen foundation models for zero-shot grounding without task-specific training. Addressing the unique geometric characteristics of remote sensing, DSEVG [49] explicitly models orientation-aware features to resolve ambiguities in direction-sensitive objects. Emerging models like MaRS [50] aim to unify representations across scales, capturing both global context and local details in very-high-resolution imagery. Overall, foundation models mainly target general understanding, while

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Find a baseball field located approximately 163 meters to the east of a building.

Find a ground track field located approximately 2 meters to the north of a building. The building is located approximately 108 meters to the northwest of a tennis court.

Find a ground track field located approximately 394 meters to the north of a building. The building is located approximately 287 meters to the south of a baseball field.

Find a pond located approximately 230 meters to the southwest of a building.

Fig. 2. Examples from the CRSVG dataset. Red boxes show the target objects found using a specific sketch and text.

TABLE I
STATISTICS OF THE CRSVG DATASET.

Split	Samples	Satellite Images	Avg Text Length
Training	14,496	9,553	15.51
Validation	3,109	2,010	15.57
Test	3,152	2,038	15.69
Total	20,757	13,601	15.55

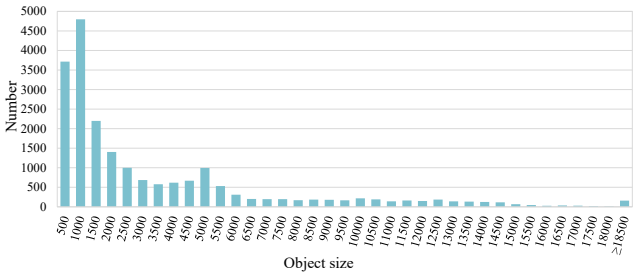


Fig. 3. Distribution of object sizes (in m²) in the CRSVG dataset.

existing grounding methods still rely primarily on text queries.

Despite these advances, most existing methods rely solely on textual descriptions to guide localization, which can be ambiguous given the complex spatial relations present in remote sensing imagery. To address this limitation, we investigate composed sketch-text visual grounding, which combines text and sketches as complementary queries to enable more accurate object localization in remote sensing images.

III. THE CRSVG DATASET

A. Problem Statement

Given a remote sensing image, composed visual grounding aims to locate a specific object within the image based on a multi-modal query consisting of both text and a sketch. Formally, the task is defined as mapping a triplet $(x_{t_i}, x_{s_i}, x_{r_i})$ to a bounding box $b_i = (x_{b_i}, y_{b_i}, w_{b_i}, h_{b_i})$, where x_{t_i} is the textual description, x_{s_i} is the sketch query, and x_{r_i} is the satellite reference image. The bounding box specifies the location and size of the target object in the image.

B. Data Annotation

To construct the dataset, we follow a series of steps to gather remote sensing images, query texts, sketches, and bounding boxes of target objects. We leverage the openly accessible OpenStreetMap (OSM) to assist in this process, as it provides valuable information on ground objects, including their categories, locations, and boundaries.

First, we download satellite images from sources such as ArcGIS and Google Maps and record their latitude and longitude coordinates. The collected images feature two specific spatial resolutions: 0.5 meters per pixel and 1 meter per pixel. Next, we select nine specific categories: baseball field, pond, roundabout, tennis court, storage tank, bridge, basketball court, track field, and swimming pool, and document their bounding boxes. Third, we utilize text templates from RSVQA [51] and RSVG [2] to create detailed descriptions that cover object attributes, relationships, and other relevant details. Both long and short descriptions are generated: long descriptions are used for text-based visual grounding, while short descriptions are employed in our method. Fourth, we draw sketches based on the boundary outlines of objects from OSM. The objects are then centered and scale-normalized on a fixed-size canvas, with random scale changes and stroke noise further applied. These perturbations deliberately reduce access to precise geometric priors, encouraging the model to focus more on coarse

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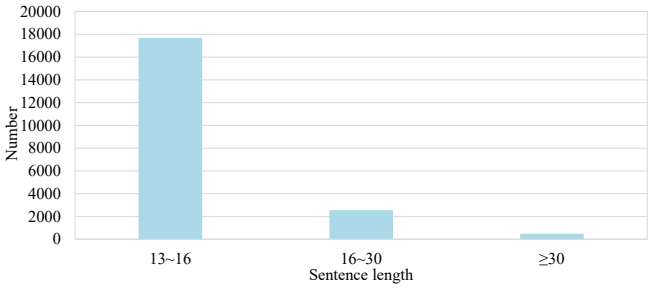


Fig. 4. Distribution of sentence lengths across the dataset.

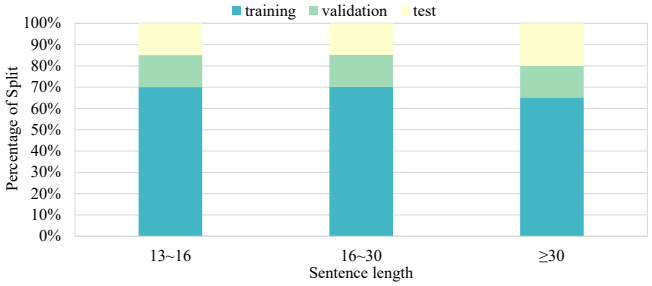


Fig. 5. Distribution of data splits by sentence length.

shapes and local spatial relationships.

Finally, to improve the annotation quality and credibility of the CRSVG dataset, we enlist experts to manually review the generated samples. The verification mainly examines the consistency among target categories, textual descriptions, bounding boxes, and sketches, as well as whether the composed query can uniquely identify the target object. Samples with ambiguous targets, inaccurate annotations, mismatched sketches, or inconsistent text-image-sketch information are corrected or discarded. Fig. 2 illustrates some representative examples of satellite imagery, sketches, text, and object bounding boxes.

C. Data Analysis

The resulting CRSVG dataset comprises 20,757 instances, each consisting of a text query, a sketch, a satellite image, and a corresponding bounding box. As summarized in Table I, these instances are derived from 13,601 unique satellite images. Specifically, the dataset features two spatial scales: 9,130 instances with a ground coverage of 256 $\times$ 256 m^2 and 11,627 instances of 512 $\times$ 512 m^2. The average length of the text queries is 15.55 words. To ensure rigorous evaluation and prevent data leakage, the dataset is partitioned into training, validation, and test sets with 14,496, 3,109, and 3,152 samples, respectively. Notably, the splitting is performed at the image level to ensure that all queries associated with the same satellite image are assigned to the same subset. The test set consists of 3,119 OSM-based samples and an additional 33 samples with manually drawn sketches, which are used specifically to evaluate robustness to human-drawn sketches. Unless otherwise specified, the following statistical analyses are based on the OSM-based samples, which serve as the primary benchmark for evaluation.

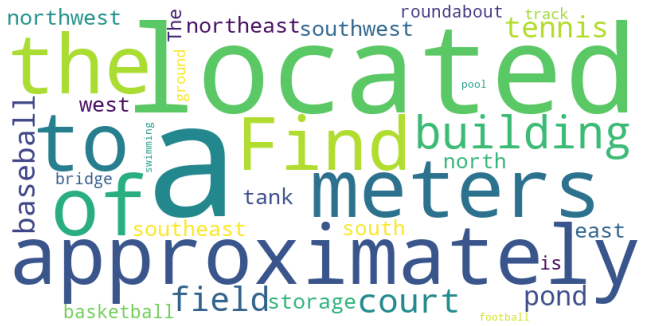


Fig. 6. Frequently occurring words in the dataset descriptions.

Fig. 3 shows the distribution of object sizes (in m^2) across the dataset. Object areas were grouped into bins with a step size of 500 m^2, ranging from 500 m^2 up to a cumulative bin of $\geq$18,500 m^2. The distribution peaks at the 1,000 m^2 interval, which contains the largest number of instances, followed by the 500 m^2 interval. This indicates that our dataset covers a wide range of object sizes. Beyond 7,000 m^2, the number of instances decreases steadily as object size increases, which is consistent with real-world observations, as very large objects are relatively rare.

Fig. 4 presents the distribution of sentence lengths in the dataset. Most instances fall within the 13–16 word range, totaling 17,690 samples, followed by the 16–30 word range with 2,554 instances. Only a small fraction of the dataset, 480 instances, contains sentences longer than 30 words. As shown in Fig. 5, the consistent distribution of sentence lengths across all dataset splits indicates that our partitioning is statistically balanced. To analyze the semantic content of the dataset, Fig. 6 displays the most frequently occurring words in the descriptions. Words such as "located", "meters", and "approximately" appear with high frequency, while spatial directions like "north" and "south", as well as object-related terms such as "field", "building", and "pond", are also commonly used.

IV. Methodology

In this section, we introduce our proposed method for composed visual grounding in remote sensing images. First, we present its overall architecture. Then, we explain the multimodal encoder and the progressive cross-scale fusion module in the proposed method. Finally, we describe the loss function applied to train the model.

A. Overview

Composed visual grounding aims to locate an object within a satellite image using a query that integrates text and a sketch. This task involves three distinct modalities: text, sketch, and satellite imagery. As illustrated in Fig. 7, we propose a novel method called CSTVG for composed visual grounding in remote sensing images. CSTVG includes a multimodal encoder, which consists of a composed query encoder and a reference encoder, along with a progressive cross-scale fusion module and a loss function. The multimodal encoder extracts features from text, sketches, and satellite images. These features are

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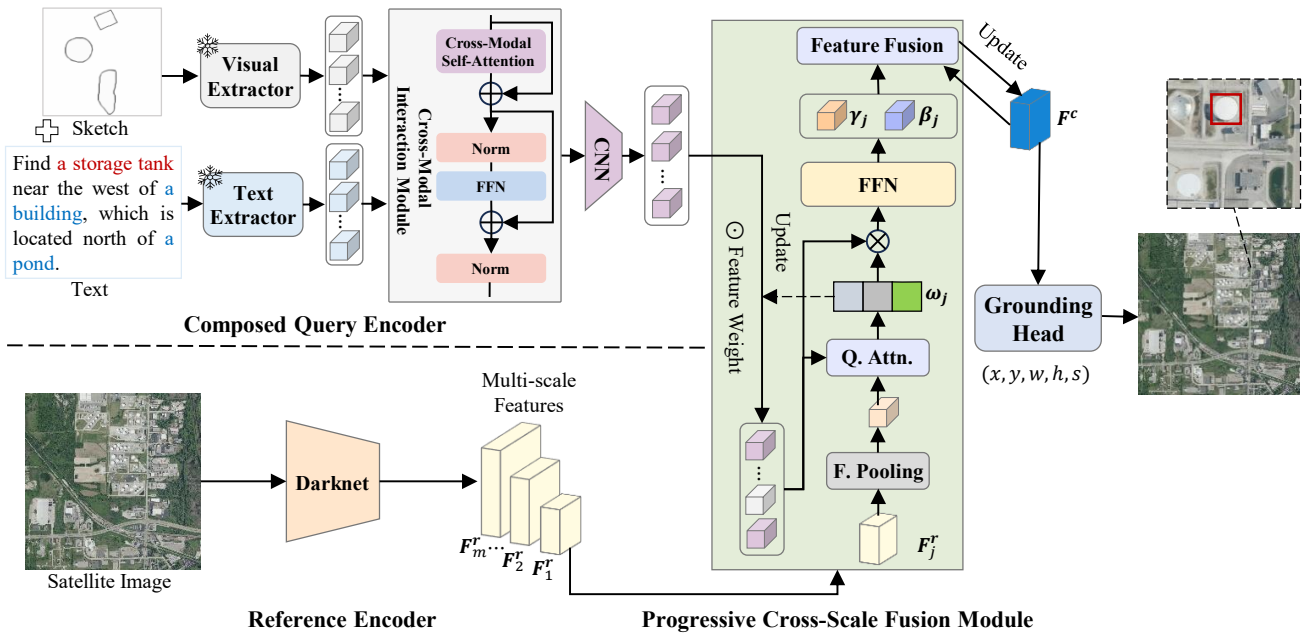


Fig. 7. Overview of the proposed CSTVG method for composed visual grounding. The framework integrates a multimodal encoder, consisting of a composed query encoder and a reference encoder, along with a progressive cross-scale fusion module.

then integrated by the fusion module to generate a grounding head, which is used to predict the object's location and size. The loss function is used to optimize the model, improving the accuracy of these predictions.

B. Multimodal Encoder

The multimodal encoder extracts features from the input data, which includes a composed query (comprising a sketch and text) and reference images (satellite images). It consists of two main components: a composed query encoder that processes the sketch and text, and a reference encoder that handles the satellite images.

Composed Query Encoder. Text and sketch belong to different modalities and exhibit a significant semantic gap. To effectively represent composed queries, it is crucial to accurately align these modalities and facilitate interaction, enabling a better understanding of the complementary information they provide. In this paper, we leverage a pre-trained CLIP model [12] and design a cross-modal interaction module to implement a composed query encoder. CLIP, a vision-language model trained on massive text-image pairs, excels at aligning image and text features. As illustrated in Fig. 7, we use the image and text encoders of the frozen CLIP model as the sketch and text feature extractors, respectively. These encoders generate sketch and text tokens, allowing for a more precise representation of both modalities.

The sketch is first resized to a fixed dimension of 224 $\times$ 224 pixels, after which it is divided into a grid of non-overlapping image patches. These patches are then processed by the second-to-last layer of the CLIP image encoder to generate sketch tokens, denoted as $\mathbf{V^s} \in \mathbb{R}^{L_s \times D}$. In this notation, each token corresponds to a specific region of the sketch, with L_s representing the length of sketch tokens and D denoting the dimension of each token. Similarly, text tokens, represented as $\mathbf{T} \in \mathbb{R}^{L_t \times D}$, are generated, where each token corresponds to a tokenized textual unit. Here, L_t is the length of the text tokens, and D is their dimensionality.

Sketches and text offer different yet complementary information. Specifically, the text tokens mainly encode semantic information about the target, such as its category, attributes, and semantic relationships, while the sketch tokens capture complementary geometric and spatial information about the target structure and configuration. To construct a unified composed query representation, we first concatenate the sketch and text tokens and then model their cross-modal interactions through self-attention, allowing information from the two modalities to be jointly integrated. To fully capture the query information and generate effective query features, we design a cross-modal interaction module. First, the sketch and text tokens are concatenated to generate cross-modal tokens, denoted as $\hat{F}^q = [V^s; T] \in \mathbb{R}^{L_q' \times D}$, where $L_q' = L_s + L_t$. Next, we employ cross-modal self-attention to compute the query features, as defined as follows:

$$\begin{aligned} \mathbf{F}' &= \mathbf{LN}(\mathbf{MHA}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) \oplus \hat{\mathbf{F}}^{\mathbf{q}}), \\ \tilde{\mathbf{F}} &= \mathbf{LN}(\mathbf{FFN}(\mathbf{F}') \oplus \mathbf{F}'), \end{aligned} \tag{1}$$

where $\mathbf{Q} = \mathbf{W_q}\hat{\mathbf{F}}^{\mathbf{q}}$, $\mathbf{K} = \mathbf{W_k}\hat{\mathbf{F}}^{\mathbf{q}}$, and $\mathbf{V} = \mathbf{W_v}\hat{\mathbf{F}}^{\mathbf{q}}$ represent the queries, keys, and values, respectively, each obtained through their corresponding learnable weight matrices $\mathbf{W_Q}$, $\mathbf{W_K}$, and $\mathbf{W_V}$. The terms $\mathbf{MHA}$, $\mathbf{LN}$, and $\mathbf{FFN}$ refer to multi-head self-attention, layer normalization, and feed-forward network, respectively. The symbol $\oplus$ represents element-wise addition. Finally, we employ a convolutional network to reduce the dimensionality of $\tilde{\mathbf{F}}$, producing the final query features $\mathbf{F^q} \in \mathbb{R}^{L^q \times D}$, which integrate information from both the sketch and the text.

Reference Encoder. Composed visual grounding aims to predict the location and size of a target object within a

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reference image, typically a satellite or aerial image. However, reference images often contain numerous objects with varying sizes, making accurate localization challenging. As highlighted in [52], [53], Darknet [54] can extract multi-scale visual features from satellite images, which is effective for representing objects at different spatial scales. Therefore, we adopt Darknet as the reference encoder to extract hierarchical visual features from the reference image.

Given a reference image $\mathbf{X}^r$, Darknet first produces a set of multi-scale feature maps:

$$\{\mathbf{G}_j^r\}_{j=1}^{S} = \mathrm{Darknet}(\mathbf{X}^r), \tag{2}$$

where $\mathbf{G}_j^r$ denotes the feature map at the j-th scale and S is the number of feature scales. The feature maps at different scales have different spatial resolutions and channel dimensions. To obtain a consistent feature representation for subsequent fusion, each scale is processed by a convolutional projection block to unify its channel dimension while retaining its scale-specific spatial resolution:

$$\mathbf{F}_j^r = \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv}_j(\mathbf{G}_j^r))), \quad j = 1, \ldots, S. \tag{3}$$

Accordingly, the reference features are represented as

$$\mathbf{F}^r = \{\mathbf{F}_j^r\}_{j=1}^{S}, \tag{4}$$

where $\mathbf{F}_j^r \in \mathbb{R}^{D \times H_j \times W_j}$, D denotes the unified channel dimension, and H_j and W_j denote the spatial height and width of the feature map at the j-th scale, respectively. Thus, $\mathbf{F}^r$ preserves the multi-scale spatial representations extracted by Darknet while providing a consistent channel dimension across scales. These features are subsequently fed into the progressive cross-scale fusion module for query-guided localization.

C. Progressive Cross-Scale Fusion Module

Satellite images usually cover large geographic areas and contain numerous visually similar objects, while composed queries often include heterogeneous information, such as object categories, visual attributes, surrounding context, and spatial relationships. In practical localization, humans typically conduct the search in a coarse-to-fine manner. They first use global semantic cues to narrow down candidate regions and then rely on finer visual and spatial cues to distinguish the target from nearby distractors. Motivated by this observation, we design a progressive cross-scale fusion module to integrate query and reference information at multiple resolutions. Intuitively, low-resolution reference features provide broader contextual and category-level information, which is suitable for identifying potential target regions, whereas higher-resolution features preserve more detailed local structures and are therefore more useful for modeling fine-grained attributes and spatial relationships. Thus, different resolutions are expected to activate different components of the query according to their visual granularity.

The proposed module progressively injects query information into the reference features through an affine transformation. The fusion process is formulated as follows:

$$\begin{aligned} \mathbf{E}_j &= \boldsymbol{\gamma}_j \odot \mathbf{Conv}\left(\mathbf{F}^c \parallel \mathbf{F}^s\right) + \boldsymbol{\beta}_j, \\ \mathbf{F}^c &= \mathbf{ReLU}\left(\mathbf{E}_j\right) + \mathbf{F}^c, \end{aligned} \tag{5}$$

where $\mathbf{F}^c$ denotes the fused feature that is progressively updated throughout the fusion process, with the corresponding reference feature serving as its initial representation. Note that $\mathbf{F}^c$ is updated iteratively across fusion steps. $\boldsymbol{\gamma}_j \in \mathbb{R}^{D \times H_c \times W_c}$ and $\boldsymbol{\beta}_j \in \mathbb{R}^{D \times H_c \times W_c}$ are affine parameters generated from the query feature $\mathbf{F}^q$. $\odot$ denotes the Hadamard element-wise product, and $\parallel$ represents feature concatenation. $\mathbf{F}^s$ denotes the spatial feature, including latitude, longitude, and ground sample distance.

Concretely, $\mathbf{F}^s \in \mathbb{R}^{10 \times H_c \times W_c}$ is a fixed, non-learnable spatial encoding defined over the $H_c \times W_c$ spatial grid of $\mathbf{F}^c$ and shared across all progressive fusion steps. The latitude- and longitude-related dimensions encode the normalized relative spatial positions of each grid cell. For the grid cell at row u and column v ($0 \le u < H_c$, $0 \le v < W_c$), $\mathbf{F}^s$ is defined as

$$\mathbf{F}^s(u,v) = \begin{bmatrix} \xi_{\min}(v),\ \phi_{\min}(u),\ \xi_{\max}(v),\ \phi_{\max}(u), \\ \xi_{\mathrm{ctr}}(v),\ \phi_{\mathrm{ctr}}(u),\ 1/H_c,\ 1/W_c,\ g/H_c,\ g/W_c \end{bmatrix}, \tag{6}$$

where ξ and ϕ denote the normalized relative coordinates along the horizontal and vertical directions of the satellite tile, respectively. The cell boundaries and centers are given by

$$\begin{aligned} \xi_{\min}(v) &= \frac{2v - W_c}{W_c}, & \xi_{\max}(v) &= \frac{2(v+1) - W_c}{W_c}, \\ \phi_{\min}(u) &= \frac{2u - H_c}{H_c}, & \phi_{\max}(u) &= \frac{2(u+1) - H_c}{H_c}, \\ \xi_{\mathrm{ctr}}(v) &= \frac{\xi_{\min}(v) + \xi_{\max}(v)}{2}, & \phi_{\mathrm{ctr}}(u) &= \frac{\phi_{\min}(u) + \phi_{\max}(u)}{2}. \end{aligned} \tag{7}$$

The first six channels encode the normalized relative position and spatial extent of each grid cell, while the next two describe the grid-cell size relative to the tile extent. The final two channels encode the spatial resolution g, i.e., the ground sample distance, as constant maps over the spatial grid. In Eq. (5), $\mathbf{F}^s$ is concatenated with the progressively updated $\mathbf{F}^c$ at each fusion step, ensuring that these positional and scale cues remain consistent throughout the modulation process. Through this affine modulation, the query information is progressively incorporated into the fused feature $\mathbf{F}^c$.

At each fusion step, the module adaptively determines which components of the query to emphasize based on the reference features at the current resolution. Specifically, $\mathbf{F}_j^r \in \mathbb{R}^{D \times H_j \times W_j}$ denotes the reference feature at the j-th resolution, where $j \in [1, \ldots, S]$ and a larger j corresponds to a higher resolution. To generate the query-weighting parameters for affine modulation, we construct the reference-context representation $\mathbf{Z}_r$ and the fused-context representation $\mathbf{Z}_c$.

The representation $\mathbf{Z}_r$ is obtained by applying average pooling to the current reference feature $\mathbf{F}_j^r$ and then repeating it along the query dimension. Therefore, $\mathbf{Z}_r$ encodes the global visual context of the reference image at the current resolution and provides resolution-dependent visual guidance for query modulation. In contrast, $\mathbf{Z}_c$ is computed from the current fused feature $\mathbf{F}^c$, which contains the accumulated cross-modal information obtained from previous fusion steps. Thus, $\mathbf{Z}_c$ reflects the current fusion state and helps determine which

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query components still need to be emphasized in the subsequent fusion process. The aligned contextual representations are then combined with the query feature to compute the query-weighting parameter:

$$\begin{aligned}\mathbf{Z}_r &= \mathbf{Rep}\left(\mathbf{AvgPool}\left(\mathbf{F}_j^r\right)\right),\\ \mathbf{Z}_c &= \mathbf{Rep}\left(\mathbf{AvgPool}\left(\mathbf{F}^c\right)\right),\\ \boldsymbol{\omega}_j &= \mathbf{softmax}\left(\frac{\mathbf{Conv}\left(\mathbf{Z}_r \odot \mathbf{Z}_c \odot \mathbf{F}^q \odot (1-\boldsymbol{\omega}_{j-1})\right)}{\boldsymbol{\tau}}\right),\end{aligned} \tag{8}$$

where $\mathbf{Rep}$ denotes the repetition operator, $\boldsymbol{\omega}_j$ is the learned query-weighting parameter at the j-th fusion step, and $\boldsymbol{\omega}_0$ is initialized to 0. $\boldsymbol{\tau}$ is a temperature factor. In this formulation, $\mathbf{Z}_r$ provides resolution-specific visual cues from the reference image, whereas $\mathbf{Z}_c$ provides the accumulated fusion context from previous steps. Their element-wise interaction with the query feature $\mathbf{F}^q$ enables the model to generate adaptive affine modulation weights conditioned on both the current visual resolution and the existing cross-modal fusion state. The term $(1-\boldsymbol{\omega}_{j-1})$ further suppresses query components that have already been strongly activated in previous fusion steps, encouraging subsequent resolutions to focus on complementary query information. Therefore, the weighting mechanism provides an intuitive way to associate low-resolution features with coarse semantic cues and high-resolution features with more detailed visual or spatial cues.

Based on the learned query-weighting parameter, we generate the affine parameters as follows:

$$\boldsymbol{\gamma}_j, \boldsymbol{\beta}_j = \mathbf{Tanh}\left(\mathbf{Rep}\left(\mathbf{FFN}\left(\mathbf{F}^q \odot \boldsymbol{\omega}_j\right)\right)\right). \tag{9}$$

The generated affine parameters are then used to update $\mathbf{F}^c$ according to Eq. (5). Since $\boldsymbol{\omega}_j$ is obtained by jointly considering the current-resolution reference feature, the fused feature, and the query feature, the affine parameters encode resolution-aware query information. Consequently, the module progressively integrates reference and query features from coarse to fine, enabling the final fused feature to concentrate more effectively on the target object.

Finally, we apply a convolutional network to the fused feature to generate the grounding head feature $\hat{\mathbf{F}}^c \in \mathbb{R}^{AK\times H_d\times W_d}$. Here, $A=9$ represents the number of predefined anchors, and $K=5$ corresponds to the parameters (x, y, w, h, s). The first four parameters denote the bounding box offsets relative to the anchors, while the last parameter s indicates the confidence score, which reflects the likelihood of an object being present at that location.

D. Loss Function

Each anchor-grid candidate in the fused feature map $\hat{\mathbf{F}}^c$ is treated as a potential target location. For each candidate, CSTVG predicts a confidence score and a set of bounding-box offsets. The confidence score indicates the likelihood that the candidate corresponds to the target object, while the offsets are used to regress its bounding box. Following [25], the overall loss function for optimizing CSTVG is defined as

$$\mathcal{L} = \mathcal{L}_{loc} + \lambda \mathcal{L}_{reg}, \tag{10}$$

where $\mathcal{L}_{loc}$ and $\mathcal{L}_{reg}$ denote the localization and bounding-box regression losses, respectively, and λ balances the two terms.

For localization supervision, one positive candidate is assigned to each training sample. It corresponds to the grid cell containing the target center and the anchor with the highest intersection over union with the ground-truth box. All remaining candidates are treated as negatives. The localization loss is defined using binary cross-entropy:

$$\mathcal{L}_{loc} = -\frac{1}{N}\sum_{i=1}^{N}\left[p_i \log \hat{p}_i + (1-p_i)\log(1-\hat{p}_i)\right], \tag{11}$$

where N is the total number of anchor-grid candidates, $\hat{p}_i$ is the confidence score of the i-th candidate, and $p_i \in \{0,1\}$ is its ground-truth label.

For bounding-box regression, we adopt predefined anchors and optimize the encoded offsets only for the positive candidate. For the k-th training sample, let (x_k, y_k, w_k, h_k) denote the center coordinates, width, and height of the ground-truth box in feature-map coordinates, $(c_{x,k}, c_{y,k})$ the coordinates of the corresponding grid cell, and $(w_{a,k}, h_{a,k})$ the dimensions of the matched anchor. Here, $c_{x,k} = \lfloor x_k \rfloor$ and $c_{y,k} = \lfloor y_k \rfloor$. The regression targets are defined as $t_{x,k} = x_k - c_{x,k}$, $t_{y,k} = y_k - c_{y,k}$, $t_{w,k} = \log(w_k/w_{a,k})$, and $t_{h,k} = \log(h_k/h_{a,k})$. Given the predicted offsets $(\hat{t}_{x,k}, \hat{t}_{y,k}, \hat{t}_{w,k}, \hat{t}_{h,k})$, the regression loss over a batch of n samples is defined as

$$\begin{aligned}\mathcal{L}_{reg} = \frac{1}{n}\sum_{k=1}^{n}\Big[&\left(\sigma(\hat{t}_{x,k}) - t_{x,k}\right)^2 + \left(\sigma(\hat{t}_{y,k}) - t_{y,k}\right)^2\\ &+ \left(\hat{t}_{w,k} - t_{w,k}\right)^2 + \left(\hat{t}_{h,k} - t_{h,k}\right)^2\Big],\end{aligned} \tag{12}$$

where $\sigma(\cdot)$ is the sigmoid function that constrains the predicted center offsets to the corresponding grid cell. During inference, the predicted center offsets are combined with the grid-cell coordinates, while the predicted scale offsets are decoded relative to the dimensions of the matched anchor to recover the final bounding box.

V. Experiments

A. Experimental Settings

*a) **Implementation Details**:* We utilize the pretrained ViT-B/32 CLIP model [12] as the backbone to extract features from both sketches and text. Following LBYLNet [26] and ReSC [25], we use the pre-trained Darknet-53 model to extract features from satellite images. The training process spans 50 epochs with a batch size of 48, using RMSprop [55] as the optimizer. We set the initial learning rate to 10^{-4} with a weight decay of 5×10^{-4}. The learning rate is reduced by half every 25 epochs. Training is conducted on two RTX 6000 GPUs, and λ is empirically set to 5.

*b) **Evaluation Metrics**:* Following established protocols [28], [56], [57], we evaluate model performance using the Intersection over Union (IoU). Given a predicted bounding box b_i and its corresponding ground-truth box b_i^*, the IoU is defined as:

$$\text{IoU}(b_i, b_i^*) = \frac{|b_i \cap b_i^*|}{|b_i \cup b_i^*|}, \tag{13}$$

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TABLE II
COMPARISON OF CSTVG WITH BASELINES ON CRSVG. BOLD INDICATES BEST PERFORMANCE(%).

Method	Test					Validation				
	acc@0.25	acc@0.50	acc@0.75	mIoU	cIoU	acc@0.25	acc@0.50	acc@0.75	mIoU	cIoU
FAOA [24]	35.08	30.62	19.85	26.26	22.22	34.35	30.72	18.78	26.04	23.75
ReSC [25]	44.28	37.61	22.89	32.34	32.02	46.70	40.66	22.74	33.82	33.64
TransVG [27]	43.85	33.08	12.31	29.30	35.18	42.42	29.31	11.05	27.93	31.58
VLTVG [28]	47.32	33.73	12.86	30.97	35.31	47.06	33.77	12.13	30.84	35.64
GeoVG [2]	56.62	52.52	37.00	44.02	43.76	56.48	51.62	34.22	43.17	45.13
RINet [31]	37.38	32.09	17.15	26.79	24.25	38.73	32.52	16.63	27.41	26.11
LQVG [32]	56.33	51.49	31.32	42.16	46.99	56.51	52.24	30.78	42.23	46.66
CSTVG	**63.99**	**60.66**	**42.42**	**49.89**	**55.20**	**64.43**	**61.18**	**41.56**	**50.08**	**54.96**

where $|\cdot|$ denotes the area of the region.

Based on this metric, we employ five indicators for a comprehensive evaluation: accuracy at specific thresholds (acc@0.25, acc@0.5, acc@0.75), mean IoU, and cumulative IoU (cIoU). The accuracy metrics, denoted as acc@t, quantify the proportion of predictions achieving an IoU score no less than a threshold t:

$$\text{acc@}t = \frac{1}{N}\sum_{i=1}^{N}\mathbb{I}\left(\text{IoU}(b_i, b_i^*) \geq t\right), \tag{14}$$

where $\mathbb{I}(\cdot)$ is the indicator function that yields 1 if the condition holds and 0 otherwise. We report results for thresholds $t \in \{0.25, 0.5, 0.75\}$. Among them, $t = 0.5$ is the standard criterion for successful localization, following common practice in visual grounding [28], [56], while $t = 0.25$ and $t = 0.75$ evaluate coarse and more precise localization, respectively.

To assess overall localization quality without relying on binary thresholding, following [58], we additionally compute:

- **Mean IoU (mIoU)**: The arithmetic mean of IoU scores across all samples, reflecting the average per-sample overlap:

$$\text{mIoU} = \frac{1}{N}\sum_{i=1}^{N}\text{IoU}(b_i, b_i^*). \tag{15}$$

- **Cumulative IoU (cIoU)**: The ratio of the total intersection area to the total union area across the entire dataset. This metric emphasizes the global overlap proportion:

$$\text{cIoU} = \frac{\sum_{i=1}^{N}|b_i \cap b_i^*|}{\sum_{i=1}^{N}|b_i \cup b_i^*|}. \tag{16}$$

Here, N represents the total number of image-query pairs in the dataset. While mIoU treats each sample equally, cIoU offers a global perspective where larger objects or predictions with higher overlap contribute more significantly to the final score.

B. Comparison with State-of-the-Art

We conduct a comprehensive comparison of our proposed method against several state-of-the-art one-stage visual grounding approaches, specifically tailored for remote sensing imagery. These baselines include FAOA [24], ReSC [25], TransVG [27], VLTVG [28], GeoVG [2], RINet [31], and LQVG [32]. Following the consensus in recent literature [2], one-stage architectures are increasingly favored in remote sensing visual grounding due to their superior inference efficiency and ability to perform end-to-end localization without the latency of region proposal generation. The specific characteristics of these comparative methods are summarized as follows:

- **FAOA** [24] proposes a fast and accurate one-stage framework that directly regresses target bounding boxes from global features, bypassing the need for two-stage region proposals, and thereby significantly accelerating inference while maintaining competitive accuracy.
- **ReSC** [25] introduces a semantic context refining module that iteratively aligns fine-grained linguistic phrases with visual regions, thereby enhancing the model's capacity to comprehend complex attributes and subtle semantic relationships.
- **TransVG** [27] leverages a pure Transformer architecture to model global interactions between image patches and query tokens, effectively capturing long-range dependencies and cross-modal correlations for precise object localization.
- **VLTVG** [28] designs a vision-language Transformer equipped with multi-granularity fusion mechanisms, enabling deep integration of visual and textual features to boost performance on challenging grounding benchmarks with diverse object scales.
- **GeoVG** [2] incorporates explicit geometric reasoning modules into the grounding pipeline, allowing the model to better interpret spatial relationships, relative positions, and directional cues described in natural language queries.
- **RINet** [31] features a region-interaction network that dynamically models the contextual relationships between candidate objects and their surroundings, improving localization robustness in dense and cluttered remote sensing scenes.
- **LQVG** [32] focuses on language query refinement by decomposing complex expressions into structural components, facilitating more precise alignment between syntactic structures and visual regions for improved grounding accuracy.

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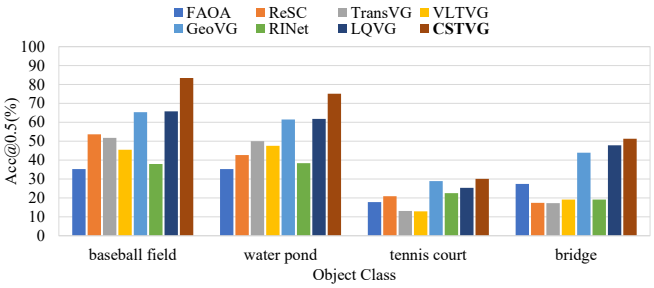


Fig. 8. Comparison of performance across different object classes on the test set.

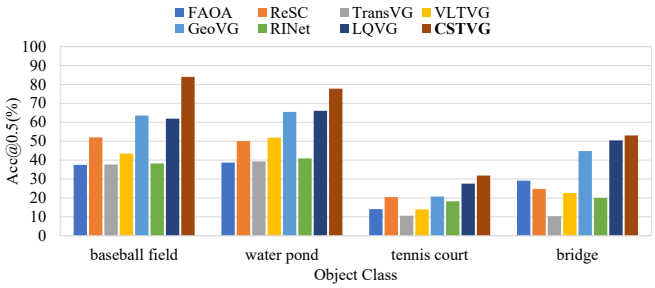


Fig. 9. Comparison of performance across different object classes on the validation set.

*a) **Performance of Visual Grounding Models**:* Table II presents a quantitative comparison between our proposed CSTVG and state-of-the-art baselines on the CRSVG dataset. We evaluate performance on both the validation and test sets using five comprehensive metrics, including accuracy at different IoU thresholds (acc@0.25, acc@0.5, acc@0.75), mean IoU (mIoU), and cumulative IoU (cIoU). From these results, three key observations can be made.

First, CSTVG consistently outperforms all baseline models across different IoU thresholds. For example, under the stringent IoU threshold acc@0.75, CSTVG achieves an absolute improvement of 5.42% over GeoVG. At the lower thresholds, acc@0.25 and acc@0.5, it surpasses GeoVG by 7.37% and 8.14%, respectively. These results indicate that CSTVG is robust under varying localization precision requirements and achieves superior cross-modal alignment. Furthermore, although performance naturally decreases as the IoU threshold increases and the task becomes more challenging, our method consistently remains ahead of all baselines. This demonstrates that CSTVG maintains stability under high-precision requirements and effectively captures fine-grained spatial features, reducing localization bias.

Second, CSTVG consistently achieves higher mIoU than the compared methods, indicating more reliable localization across individual objects. Evaluating mIoU allows us to measure the model's average performance on instances of varying scales and aspect ratios. For example, our model attains 49.89%, outperforming GeoVG by 5.87%, which demonstrates that CSTVG produces more balanced and accurate predictions while effectively mitigating the localization bias common in existing methods.

Third, CSTVG achieves a significantly higher cIoU compared to existing methods. On the test dataset, it achieves a cIoU of 55.20%, surpassing the baseline model LQVG by 8.21%. The high cIoU confirms that CSTVG provides excellent global overlap across the entire dataset, indicating that the predicted bounding boxes are well aligned with the ground-truth annotations. This superior performance is attributed to our progressive cross-scale fusion module, which gradually integrates text and sketch cues with visual features, enabling the model to capture the global scene structure without sacrificing local details.

*b) **Performance with Different Categories**:* To evaluate the performance of our model against existing baselines, we selected four representative categories for comparative analysis. The results on the test and validation sets are illustrated in Fig. 8 and Fig. 9, respectively, with acc@0.5 employed as the primary metric. Focusing on the test set, it can be observed that CSTVG consistently outperforms all baseline models across all selected categories. For example, for targets such as baseball fields and water ponds, our model achieves a clear advantage, with accuracies of 83.41% and 75.11%, respectively. For more challenging categories, such as tennis courts and bridges, the overall performance decreases due to complex backgrounds and varying object scales. Nevertheless, CSTVG consistently maintains its advantage. Notably, the performance trends on the validation set closely mirror those on the test set, further highlighting the strong adaptability and robustness of our method in handling diverse remote sensing targets.

TABLE III
PERFORMANCE (ACC@0.5) ON DIFFERENT OBJECT SIZES (m^2).

Set	Method	1,000	2,000	3,000	4,000	≥4,000
Test	FAOA [24]	22.36	26.35	38.35	33.86	36.62
	ReSC [25]	22.72	30.33	42.05	35.45	53.01
	TransVG [27]	18.84	20.47	45.00	27.27	50.00
	VLTVG [28]	17.17	24.74	36.65	42.86	50.31
	GeoVG [2]	35.42	46.35	56.82	58.73	66.39
	RINet [31]	20.82	29.59	38.03	32.67	40.80
	LQVG [32]	34.52	44.39	61.13	58.42	67.17
	CSTVG	**36.67**	**51.18**	**62.22**	**76.19**	**81.33**
Val	FAOA [24]	17.79	27.80	34.65	34.16	39.61
	ReSC [25]	24.20	34.97	44.23	42.08	55.44
	TransVG [27]	11.59	20.61	36.84	27.27	45.74
	VLTVG [28]	15.66	25.64	40.85	43.56	49.19
	GeoVG [2]	29.36	44.67	58.87	63.37	67.71
	RINet [31]	22.36	28.06	35.80	38.10	39.63
	LQVG [32]	31.66	43.41	53.98	57.67	69.71
	CSTVG	**33.63**	**50.05**	**74.65**	**73.76**	**82.67**

*c) **Performance with Different Object Sizes**:* To verify the robustness of CSTVG across diverse object sizes, we categorized objects into five size intervals (in m^2): [0, 1000), [1000, 2000), [2000, 3000), [3000, 4000), and $\geq$ 4000. The comparative results in Table III show that our method consistently achieves the highest accuracy across all categories on both the test and validation sets.

Specifically, for small objects in the [0, 1000) m^2 range,

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TABLE IV
PERFORMANCE (ACC@0.5) COMPARISON ACROSS DIFFERENT QUERY LENGTH GROUPS.

Set	Method	13–16	16–30	≥30
Test	FAOA [24]	30.98	28.50	29.17
	ReSC [25]	36.76	41.95	43.75
	TransVG [27]	32.20	39.13	33.33
	VLTVG [28]	34.76	29.82	20.83
	GeoVG [2]	52.16	55.15	52.08
	RINet [31]	32.30	31.93	27.08
	LQVG [32]	50.72	56.73	52.08
	CSTVG	**60.36**	**61.74**	**64.58**
Val	FAOA [24]	31.53	26.89	21.13
	ReSC [25]	40.56	40.47	45.07
	TransVG [27]	28.13	37.04	25.00
	VLTVG [28]	34.39	30.55	28.17
	GeoVG [2]	51.56	54.05	40.85
	RINet [31]	33.37	28.20	23.94
	LQVG [32]	52.17	53.79	46.48
	CSTVG	**61.62**	**59.27**	**54.93**

TABLE V
QUANTITATIVE COMPARISON OF CSTVG WITH OSM-BASED AND HUMAN-DRAWN SKETCH INPUTS.

Sketch Input	acc@0.25	acc@0.50	acc@0.75
OSM-based	63.64	60.61	42.42
Human-drawn	60.61	57.58	39.39

CSTVG achieves an accuracy of 36.67% on the test set, outperforming all baseline models. This highlights its capability to localize compact targets. For large-scale objects ($\geq 4000\ m^2$), the model attains a notable accuracy of 81.33%, exceeding the state-of-the-art model LQVG by 14.16%. Moreover, as object size increases, the performance gap between CSTVG and other methods widens progressively, indicating that our approach effectively captures global contextual information while preserving local spatial coherence. These results demonstrate that CSTVG successfully bridges local detail preservation and global structure understanding, maintaining stable performance across extreme scale variations.

*d) **Performance with Different Sentence Lengths**:* To evaluate the model's adaptability to textual complexity, we divided the samples based on the length of their short text queries into three intervals: 13–16, 16–30, and ≥ 30 words. For each group, we compared CSTVG using short texts and sketches with the baseline using the corresponding long texts. The results, summarized in Table IV, show that CSTVG consistently achieves the best performance across all sentence lengths on both the test and validation sets.

Specifically, for complex descriptions of 30 words or more, CSTVG achieves an accuracy of 64.58% on the test set, outperforming the best baseline model LQVG by 12.5%. A similar trend is observed for queries of 16 to 30 words, where CSTVG leads LQVG by 5.01%. Even for concise queries of 13 to 16 words, our model maintains a substantial margin of 8.2% over the state-of-the-art baseline. These results indicate that, while traditional methods often struggle to parse dense linguistic information, CSTVG effectively captures long-range dependencies and, by leveraging sketches, accurately maps complex spatial relations to the corresponding geographic regions. The stable performance across varying sentence lengths highlights the robustness of our framework in handling different levels of descriptive complexity in remote sensing scenes.

TABLE VI
COMPARISON OF MODEL COMPLEXITY AND INFERENCE EFFICIENCY.

Method	Params (M) ↓	Time (ms) ↓	FPS ↑
TransVG [27] + Sketch	301.19	23.71	42.17
LQVG [32] + Sketch	317.85	33.57	29.79
CSTVG	**241.09**	**20.35**	**49.14**

*e) **Inference Efficiency**:* To further evaluate the efficiency of CSTVG, we compare its model size and inference speed with TransVG [27] and LQVG [32]. All models use the same multimodal input setting and are tested on a single NVIDIA RTX 6000 Ada GPU with the same batch setting. As shown in Table VI, CSTVG has fewer parameters and achieves the lowest inference time of 20.35 ms and the highest speed of 49.14 FPS. These results demonstrate that CSTVG maintains efficient inference while effectively incorporating both sketch and text information.

In summary, all the results presented above demonstrate that the integration of the sketch strategy significantly improves localization accuracy across diverse scenarios. The progressive cross-scale fusion module effectively ensures that CSTVG maintains optimal performance regardless of object scale, addressing a major challenge in remote sensing grounding. Notably, these enhancements do not sacrifice the model's linguistic comprehension. In fact, CSTVG achieves superior performance with short, medium, and long textual descriptions, outperforming all baseline models. These consistent improvements across various object categories, object sizes, and query complexities prove that our proposed framework is robust and highly effective.

C. Robustness Evaluation with Human-Drawn Sketches

To evaluate the robustness of the proposed method using more realistic sketch inputs, we construct an additional human-drawn sketch test subset. Specifically, we randomly select 33 samples from the original test set and ask human participants to draw a sketch for each sample. These human-drawn sketches are used exclusively for testing and are compared with the OSM-based sketches of the same samples in the original test set.

Table V presents a quantitative comparison of CSTVG using OSM-based and human-drawn sketches on the same test subset of 33 samples. Overall, CSTVG maintains largely consistent performance across the two sketch types, with only a slight decrease when using human-drawn sketches. Despite natural differences in drawing style, human-drawn sketches retain the key structural and spatial information of the target objects. CSTVG effectively captures these essential cues and integrates

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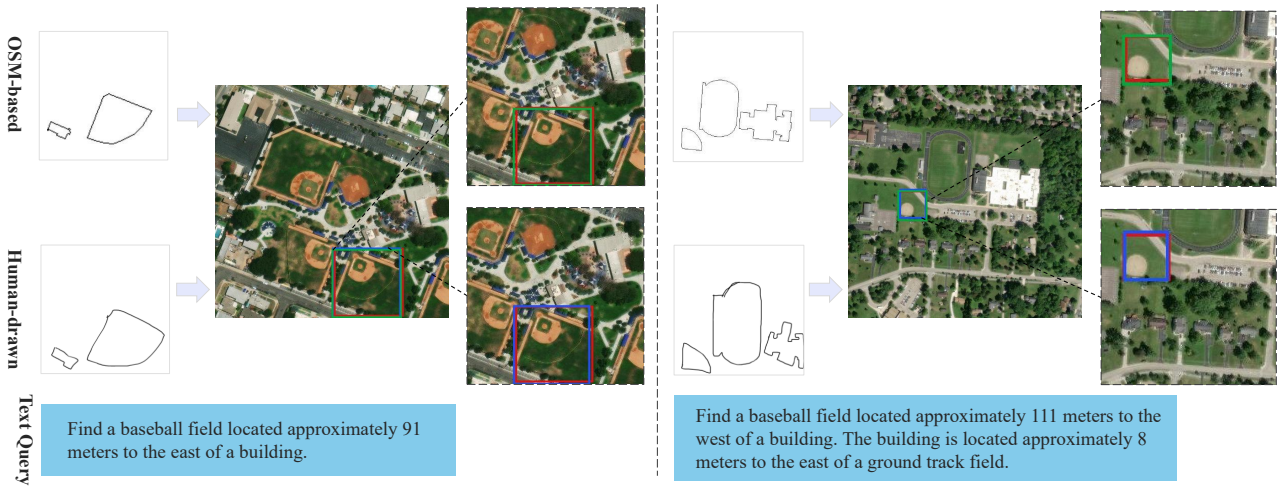


Fig. 10. Qualitative comparison of CSTVG predictions using OSM-based and human-drawn sketches. Red, green, and blue bounding boxes denote the ground truth, OSM-based predictions, and human-drawn predictions, respectively.

TABLE VII
COMPARISON WITH SKETCH-AUGMENTED GROUNDING BASELINES UNDER THE SAME MULTIMODAL INPUT SETTING.

Set	Method	acc@0.25	acc@0.50	acc@0.75
Test	TransVG [27] + Sketch	46.78	34.21	12.66
	VLTVG [28] + Sketch	49.63	42.23	22.64
	RINet [31] + Sketch	37.83	31.45	16.48
	LQVG [32] + Sketch	60.76	56.97	37.48
	CSTVG	**63.99**	**60.66**	**42.42**
Val	TransVG [27] + Sketch	47.54	35.32	13.83
	VLTVG [28] + Sketch	51.08	42.84	22.84
	RINet [31] + Sketch	38.18	32.65	15.50
	LQVG [32] + Sketch	60.79	56.80	38.21
	CSTVG	**64.43**	**61.18**	**41.56**

them with complementary textual information, demonstrating that it generalizes well from OSM-based sketches to realistic human-drawn inputs.

Fig. 10 further presents representative qualitative results. As shown, CSTVG consistently localizes the target objects across different drawing styles. These results further demonstrate its ability to effectively integrate the semantic cues from text with the structural and spatial cues from sketches, supporting its robustness to realistic human-drawn inputs.

D. Comparison with Multimodal Extensions of Existing Grounding Methods

CSTVG performs visual grounding with composed sketch-text queries, whereas most existing grounding methods are designed primarily for textual queries. To enable a more comprehensive comparison under a unified multimodal input setting, we construct sketch-augmented variants of four state-of-the-art grounding methods: TransVG [27], VLTVG [28], RINet [31], and LQVG [32]. Specifically, each method is equipped with the same sketch encoder used in CSTVG, and the resulting sketch features are incorporated into its original query representation through feature concatenation. Apart from the additional sketch branch and the corresponding feature adaptation, the original architecture of each method remains unchanged. All models are trained and evaluated using the same data splits and experimental settings. This setup enables a direct comparison under identical input conditions, with every method having access to both textual and sketch information.

Table VII presents the results of this comparison. CSTVG outperforms all sketch-augmented baselines across the three IoU thresholds on both the test and validation sets, showing that its advantage is retained when the competing methods are provided with the same sketch-text queries. Among these baselines, the sketch-augmented LQVG variant achieves the strongest overall performance. On the test set, CSTVG improves acc@0.25, acc@0.50, and acc@0.75 from 60.76%, 56.97%, and 37.48% to 63.99%, 60.66%, and 42.42%, corresponding to gains of 3.23, 3.69, and 4.94 percentage points, respectively. The same trend is observed on the validation set, where CSTVG surpasses the sketch-augmented LQVG variant by 3.64, 4.38, and 3.35 percentage points at the three thresholds.

The consistent gains across both data splits further substantiate the advantage of CSTVG. In particular, under the more stringent acc@0.75 criterion, CSTVG outperforms the strongest sketch-augmented baseline by 4.94 and 3.35 percentage points on the test and validation sets, respectively. These gains demonstrate its stronger capability for precise target localization rather than coarse region identification. Overall, these findings indicate that the superior performance of CSTVG does not arise merely from its access to sketch inputs. Rather, they validate the effectiveness of the proposed CSTVG framework, particularly its multimodal modeling and cross-modal fusion mechanisms, in jointly exploiting the complementary semantic cues from text and spatial cues from sketches.

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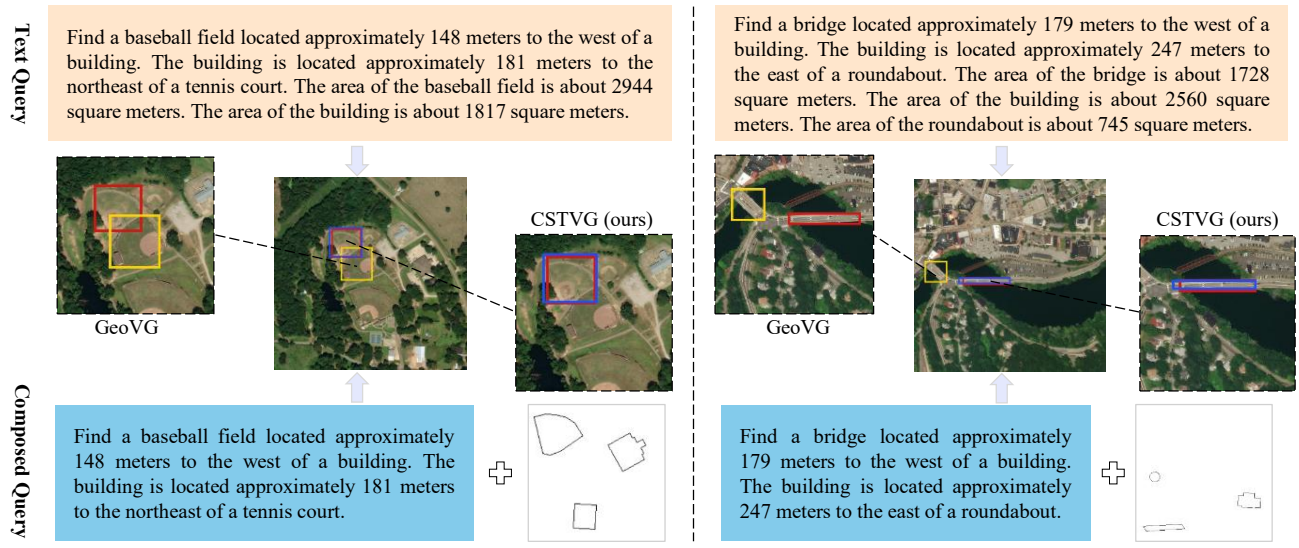


Fig. 11. Visualization comparison of prediction results for GeoVG and CSTVG. Red bounding boxes denote ground truth, blue bounding boxes represent predictions made by CSTVG, and yellow bounding boxes denote selections made by GeoVG.

E. Qualitative Analysis

We conducted a qualitative analysis of CSTVG and a baseline method (GeoVG) by comparing their predicted results, as shown in Fig. 11. In this figure, ground-truth bounding boxes are indicated in red, while blue bounding boxes represent predictions made by CSTVG and yellow bounding boxes denote selections made by GeoVG. We observed that CSTVG identifies more accurately the locations of target objects compared to GeoVG, which tends to detect non-target objects more frequently. For example, in Fig. 11, the text component of the composed query describes a building as a reference object for the target object. However, if the building's shape and its relationships with other objects are not clearly specified, the models may incorrectly identify nearby objects. Furthermore, despite detailed descriptions of spatial relationships in the query text, these descriptions rely heavily on numerical information, which current natural language models struggle to interpret accurately. Consequently, sketches, which are easier for models to understand than text descriptions of spatial relationships and shapes, provide a significant advantage. In summary, our method, which utilizes composed queries, generally achieves better accuracy than text-based visual grounding.

Additionally, to provide a more intuitive explanation of the model's localization behavior, we visualize representative grounding results together with their corresponding attention maps in Fig. 12. The examples cover four representative object categories, namely water pond, storage tank, bridge, and baseball field. Each case includes the input image with the ground-truth box in red and the predicted box in green, as well as the corresponding attention map.

Overall, the predicted bounding boxes exhibit high overlap with the ground-truth annotations across objects of varying scales, shapes, and aspect ratios, demonstrating the strong localization capability of the proposed model. As shown in Fig. 12, for regularly shaped targets such as storage tanks, the model accurately captures the circular boundaries, while for elongated structures such as bridges, the predicted boxes remain tightly aligned with the target regions. These results indicate that the model can effectively infer object extents from the query and achieve reliable spatial grounding.

The attention maps further reveal that the model learns meaningful visual focus patterns that are well aligned with object characteristics. For visually distinctive targets such as water ponds and storage tanks shown in the first row of Fig. 12, the responses are highly concentrated on the target regions, with limited activation in irrelevant background areas, indicating that the model can identify the most discriminative visual cues associated with the query. For linear structures such as bridges in the second row of Fig. 12, the attention responses extend along the dominant direction of the object, which is consistent with its geometric structure.

Moreover, in cluttered scenes, the visualizations highlight the robustness of CSTVG. For example, in the baseball field examples shown in the second row and second column of Fig. 12, even when multiple visually similar candidates or distracting surrounding structures are present, the attention maps remain focused on the correct target region. This demonstrates that CSTVG can effectively exploit fine-grained semantic cues from the query to distinguish the referred object from confusing instances and reduce incorrect detections in challenging remote-sensing scenarios.

F. Ablation Study

Effects of key components and query modalities. To evaluate the impact of different components and input data in our proposed approach, we performed an ablation study on the test and validation datasets. The accuracy results (%) are summarized in Table VIII. We specifically assessed the effects of removing the progressive cross-scale fusion module ('w/o PRFusion'), the sketch query ('w/o sketch'), and the text query ('w/o text'). 'w/o PRFusion' refers to the replacement of the progressive cross-scale fusion module with a single-step

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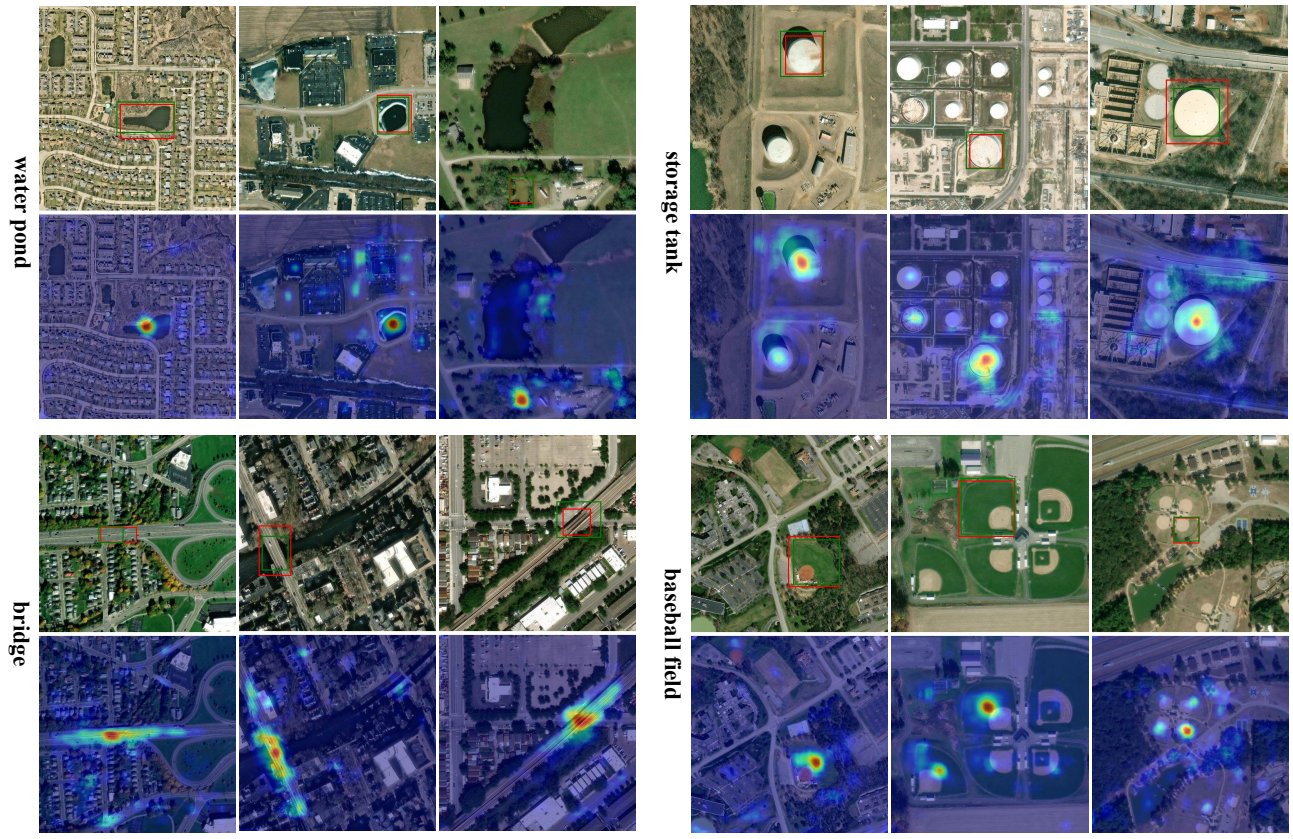


Fig. 12. Qualitative visualization of CSTVG's grounding performance and attention maps across four representative objects.

TABLE VIII
ABLATION TEST RESULTS IN TERMS OF ACCURACY(%) ARE OBTAINED BY DISCARDING DIFFERENT COMPONENTS.

Method	Test		Validation	
	acc@0.25	acc@0.50	acc@0.25	acc@0.50
w/o PRFusion	50.98	48.03	49.15	46.57
w/o sketch	55.56	50.34	53.59	49.34
w/o text	55.24	53.00	56.90	53.65
CSTVG	**63.99**	**60.66**	**64.43**	**61.18**

fusion operation, which directly combines the high-resolution reference features with the query features without progressive cross-scale fusion. 'w/o sketch' indicates the exclusion of the sketch query. Finally, 'w/o text' refers to the scenario where only the sketch query is used as input data.

The results yield three key observations. First, the progressive cross-scale fusion module significantly enhances the accuracy of CSTVG. For example, on the test dataset, CSTVG achieves an absolute improvement of 12.63% and 13.01% in acc@0.5 and acc@0.25, respectively, compared to 'w/o PRFusion'. This improvement is likely due to the fusion module's ability to integrate query features with reference features at varying resolutions, which better highlights the target objects. Second, both sketch and text data provide valuable supplementary information that boosts CSTVG's accuracy. For example, CSTVG exhibits higher accuracy in acc@0.5 on the test dataset when compared to 'w/o sketch' and 'w/o text'. Third, when only one modality (either text or sketch) is used, sketches tend to be easier for the model to interpret, resulting in more accurate target localization. For example, compared to 'w/o sketch', 'w/o text' achieves a 2.66% improvement in acc@0.5 on the test dataset.

Effect of PRFusion across visual backbones. To investigate the generalizability of PRFusion across different visual architectures, we replace the default Darknet backbone with Swin Transformer, a representative hierarchical vision Transformer capable of producing multi-scale feature representations. For each backbone, we evaluate the model both with and without PRFusion under the same training and evaluation settings. As shown in Table IX, PRFusion consistently improves localization accuracy across both backbones, all three IoU thresholds, and both data splits. On the test set, incorporating PRFusion improves the Swin Transformer backbone by 7.35, 8.69, and 6.22 percentage points at acc@0.25, acc@0.50, and acc@0.75, respectively. The corresponding improvements with Darknet are 13.01, 12.63, and 9.97 percentage points. Consistent gains are also observed on the validation set, ranging from 6.63 to 9.26 percentage points for Swin Transformer and from 9.91 to 15.28 percentage points for Darknet. These consistent improvements demonstrate that the effectiveness of PRFusion is not restricted to the default CNN backbone and that the proposed strategy can also be applied effectively to a hierarchical Transformer architecture.

Although Swin Transformer achieves lower absolute accu-

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TABLE IX
EFFECT OF PRFUSION WITH DIFFERENT VISUAL BACKBONES ON THE TEST AND VALIDATION SETS IN TERMS OF ACCURACY (%)

Set	Backbone	PRFusion	acc@0.25	acc@0.50	acc@0.75
Test	Swin Transformer	✗	42.67	32.09	12.60
		✓	50.02	40.78	18.82
	Darknet	✗	50.98	48.03	32.45
		✓	**63.99**	**60.66**	**42.42**
Val	Swin Transformer	✗	41.81	31.07	11.51
		✓	49.15	40.33	18.14
	Darknet	✗	49.15	46.57	31.65
		✓	**64.43**	**61.18**	**41.56**

racy than Darknet in the current experiments, the comparison is based on a direct backbone replacement under the same localization framework. The overall localization framework is built upon an anchor-based YOLO architecture whose multi-scale feature aggregation and prediction heads were originally configured for the convolutional features produced by Darknet. In this ablation experiment, we directly replace Darknet with Swin Transformer while keeping the remaining architecture and training settings unchanged, without introducing Swin-specific modifications to the feature aggregation module, prediction head, or optimization strategy. Therefore, the lower performance of Swin Transformer may partly result from the lack of architecture-specific adaptation and should not be interpreted as indicating an inherent limitation of Transformer backbones for this task. The purpose of this experiment is not to establish an optimal Transformer-based implementation, but to determine whether PRFusion remains effective when applied to a substantially different feature extractor. The consistent improvements obtained after introducing PRFusion confirm its applicability to both CNN- and Transformer-based backbones under the same localization framework.

G. Discussion

Composed visual grounding combines the semantic information conveyed by text with the geometric and spatial cues provided by sketches, helping reduce localization ambiguity in complex remote sensing scenes. This is particularly useful when multiple candidate objects share similar semantic attributes or when the target is mainly distinguished by its spatial configuration. In such cases, text alone may be insufficient to precisely describe relative positions, shapes, or spatial layouts. Without sketch information, these spatial constraints must be expressed entirely through text, which may increase description complexity and still leave ambiguity among similar candidate objects. By contrast, sketches can directly encode such geometric and spatial information, providing complementary constraints that help distinguish the intended target. Therefore, sketches are especially valuable when spatial structure plays a key role in target identification. This capability gives composed visual grounding potential applications in urban monitoring, disaster assessment, infrastructure inspection, and search-and-rescue. For example, during post-disaster assessment, an analyst could use a touch-enabled terminal to sketch the approximate layout of a damaged building relative to nearby roads and combine it with a brief textual description, allowing a target that is difficult to describe using language alone to be specified more intuitively.

Despite this potential, performance may be sensitive to variations in the completeness and level of abstraction of human-drawn sketches. Cross-modal alignment also remains challenging under scale variations, complex backgrounds, and diverse object appearances, while constructing large-scale benchmarks with realistic human-drawn sketches is labor-intensive. These challenges suggest several future research directions, including robust modeling of incomplete sketches with varying levels of abstraction, efficient construction of large-scale human-drawn sketch benchmarks, and foundation-model-based multimodal representation learning to improve cross-scene and cross-user generalization.

VI. CONCLUSION

In this paper, we investigate composed sketch-text visual grounding in remote sensing, which aims to locate objects in satellite images using both sketches and text. To address this problem, we propose the CSTVG method, which integrates a multimodal encoder with a progressive cross-scale fusion module. The encoder processes data from different modalities through a composed query encoder and a reference encoder. The fusion module refines the query features by progressively integrating them with reference features at varying resolutions to generate fused features. These fused features are then used to accurately predict the target objects. Additionally, we evaluate our method on the large-scale CRSVG benchmark. Extensive experiments demonstrate that CSTVG outperforms current state-of-the-art methods.

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